

Umesh Kashyap

✉ umeshk@iitbhillai.ac.in 📍 Bhilai, India [🌐 LinkedIn](#) [🐙 GitHub](#) [🔍 Google Scholar](#)

PROFILE SUMMARY

Ph.D. Scholar at the **Indian Institute of Technology (IIT) Bhilai** with 4+ years of research experience in **Trustworthy AI** and **Adversarial Machine Learning**. Specialized in developing robust defense mechanisms against adversarial attacks, perceptual privacy protection, and deepfake detection.

Proven track record of publishing in high-impact venues (IEEE Transactions, PETs, ICDAR) and securing competitive research funding (TIH Fellowship, NFOBC). Seeking a Post-Doctoral position to leverage expertise in secure deep learning architectures and contribute to cutting-edge research in AI safety.

EDUCATION

Ph.D. in Computer Science (Ongoing) 2022 - Present
Indian Institute of Technology Bhilai, India
Thesis Focus: Adversarial Security Evaluation & Privacy-Preserving Deep Learning

M.Sc. Computer Science 2018 - 2020
Atal Bihari Vajpayee University, Bilaspur, India

B.Sc. Computer Science 2015 - 2018
Bilaspur University, Bilaspur, India

RESEARCH INTERESTS

- **Adversarial Machine Learning:** Attacks (Patch, Gradient-based) and Defense mechanisms.
- **Privacy-Preserving AI:** Perceptual encryption, secure inference, and data privacy.
- **Forensics:** Deepfake detection, attribution, and style-based anomaly detection.
- **Trustworthy AI:** Robustness evaluation and model security benchmarking.

PUBLICATIONS

1. **U. Kashyap**, R. Borikar, S. K. Padhi, S. S. Ali, “*The Cosine Conspiracy: Subverting CLIPure-Cos with a Latent-Space Informed Attack.*” **International Conference on Security, Privacy, and Applied Cryptography Engineering (SPACE)**, 2026. **accepted**
2. S. K. Padhi, A. Tiwari, **U. Kashyap**, S. S. Ali, “*Doc-Protector: A Self-Healing Approach for Digital Documents.*” **The 20th International Conference on Document Analysis and Recognition (ICDAR)**, 2026.
3. **U. Kashyap**, S. S. Ali, “*GAN-Invert: Unveiling Vulnerabilities in Privacy-Preserving Facial Transformations.*” **Proceedings on Privacy Enhancing Technologies**, 2026.
4. **U. Kashyap**, A. Vishwakarma, S. S. Ali, “*Generalize Adversarial Malware Detection.*” **Journal of Hardware and Systems Security**, 10(1), p.11, 2026.
5. S. Nevase, **U. Kashyap**, S. S. Ali, “*Perceptually-Guided Adversarial Diffusion: Enhancing Visual Fidelity in Facial Privacy.*” **International Conference on Computer Vision and Image Processing (CVIP)**, 2026.
6. R. Kumar, **U. Kashyap**, S. S. Ali, “*Gradient-Guided Adversarial Patch Attack for Deep Neural Networks.*” **International Conference on Security, Privacy, and Applied Cryptography Engineering (SPACE)**, pp. 227–245, 2026.
7. **U. Kashyap**, S. K. Padhi, S. S. Ali, “*Is Perceptual Encryption Secure? A Security Benchmark for Perceptual Encryption Methods.*” **IEEE Transactions on Artificial Intelligence**, 2025.
8. S. K. Padhi, H. Kumar, **U. Kashyap**, S. S. Ali, “*De-Fake: Style-Based Anomaly Deepfake Detection.*” preprint arXiv:2507.03334, 2025.
9. A. Vishwakarma, **U. Kashyap**, S. S. Ali, “*Adversarial Malware Detection.*” **International Conference on Security, Privacy, and Applied Cryptography Engineering (SPACE)**, pp. 277–286, 2024.
10. S. M. Shahid, S. K. Padhi, **U. Kashyap**, S. S. Ali, “*Generalized Deepfake Attribution.*” preprint arXiv:2406.18278, 2024.

FELLOWSHIPS & AWARDS

- **TIH PhD Fellowship:** Awarded by the Technology Innovation Hub (TIH) for research on AI-driven GST fraud detection (Dec 2023 – May 2025).
- **National Eligibility Test (UGC-NET):** Qualified **6 times** (2019–2025), demonstrating consistent top-tier academic competency nationwide.

- **NFOBC Doctoral Fellowship:** Awarded by University Grants Commission (2023, 2024).
- **CG-SET Qualified:** Chhattisgarh State Eligibility Test (2019).

RESEARCH & PROFESSIONAL EXPERIENCE

PhD Research Scholar Jan 2022 - Present
MIST Lab, IIT Bhilai

- Led the development of a security benchmark for perceptual encryption, identifying critical vulnerabilities in existing state-of-the-art methods.
- Designed "De-Fake," a novel style-based anomaly detection framework for identifying deepfakes, improving detection rates on unseen datasets.
- Conducted extensive adversarial evaluation on malware detection systems, proposing robust countermeasures against evasion attacks.

Project Scientist Jan 2023 - Dec 2023
Indian Institute of Technology (IIT) Delhi

- Collaborated on the evaluation of next-generation wireless communication systems using ML-driven optimization techniques.
- Validated performance metrics through rigorous experimental setups and produced technical documentation for grant compliance.

Guest Faculty (Computer Science) 2020 - 2022
Atal Bihari Vajpayee University

- Taught core curriculum: Machine Learning, Data Structures, and Programming.
- Mentored undergraduate students on capstone projects focusing on applied ML.

TECHNICAL SKILLS

Languages	Python, C, C++, LaTeX, SQL
ML Framework	PyTorch, TensorFlow, Keras, Scikit-learn, Numpy, Pandas, Seaborn, Matplotlib, Open-CV
Tools & Platforms	Git, Docker, Linux, Google Colab, Jupyter, Hugging Face
Core Competencies	Model Robustness, Adversarial Attack/Defense, Computer Vision, NLP

LANGUAGES

Hindi: Native **English:** Professional